Prabhat Vilas Karlekar

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Summary

Software Engineer with 3.5+ years of experience designing distributed systems, network protocols, cryptographic infrastructures, applied ML systems, and privacy-preserving AI frameworks. Experienced in system architecture, protocol simulation, applied cryptography, and high-performance computing, with IEEE-published research.

Experience

Project Engineer – HPC Infrastructure & Ecosystem **Oct 2024 – Present**
Centre for Development of Advanced Computing (CDAC) *Pune, Maharashtra*

- Designed distributed system architectures for privacy-preserving AI, cryptographic networking, and decentralized infrastructure, emphasizing modularity, scalability, and protocol-driven communication.
- Architected PAYF, a cryptographically verifiable network-layer economic forwarding framework integrating protocol simulation, distributed settlement, and authenticated packet forwarding using ns-3 and Rust.
- Simulated PAYF across multiple network topologies and traffic conditions to validate forwarding correctness, protocol behavior, and billing verification.
- Designed and developed a privacy-preserving Federated Learning framework integrating Multi-Key CKKS Homomorphic Encryption with TensorFlow/Keras for secure distributed model aggregation across HPC nodes.
- Designed a decentralized architecture integrating blockchain marketplaces with QoS-driven resource allocation for low-latency network services.
- Authored HLDs, LLDs, IETF Internet-Draft-style protocol specifications, whitepapers, and patent-oriented Invention Disclosure Forms (IDFs) for AI, networking, and cryptographic systems.

Quality Analyst – JioCloud **Dec 2023 – Sep 2024**
Reliance Jio Platforms Limited *Navi Mumbai, Maharashtra*

- Worked closely with QA and platform teams to design and execute test scenarios, reducing manual regression testing and accelerating feature deployment cycles in Agile environments.
- Collaborated with cross-functional engineering teams, improving defect triage, root-cause analysis, and release stability by investigating defects, reproducing issues, and validating fixes across system components.

Graduate Engineer Trainee **Dec 2022 – Nov 2023**
Reliance Jio Platforms Limited *Navi Mumbai, Maharashtra*

- Gained hands-on exposure to cloud platforms, DevOps workflows, UAT, API testing and production monitoring practices, improving system stability and user experience.
- Adapted testing strategies to changing requirements in a fast-paced environment reducing issue resolution time through knowledge sharing and test case optimization.

Technical Skills

Languages: Python, C/C++, Rust, Bash/Shell
Systems & Backend: Distributed Systems, System Design, Linux, Network Protocols, ns-3, Rust/Axum, Apache Kafka, WebAssembly
AI/ML & HPC: TensorFlow, Keras, LSTM/GRU, Scikit-learn, NumPy, Pandas, Federated Learning, Time-Series Forecasting, FinBERT, MPI, HPC
Cryptography: SHA, Homomorphic Encryption, PKI, OpenFHE (CKKS), Proxy Re-Encryption, Merkle Trees, Smart Contracts
Development Tools: GitLab/GitHub, CI/CD, Pytest, Postman, Charles Proxy

Research & Projects

PAYF – Pay As You Forward | *ns-3, Rust* **Apr 2026**

- Designed and implemented a cryptographically verifiable network-layer economic framework for authenticated packet forwarding and both centralized and distributed settlement.
- Built modular networking components including signed QoS shim headers, FlowCache, forwarding/accounting logic, and settlement engines for verifiable flow-based billing.
- Integrated PostgreSQL, Rust/Axum APIs, and a Yew/WebAssembly dashboard for settlement and protocol visualization.

GreyMisnomer – Open-Source Verifiable Carbon Credit Registry SDK | Live Demo **Jan 2026**

- Designed an open-source Rust-based registry protocol for verifiable carbon-credit lifecycle management with protocol-level prevention of double counting and greenwashing.
- Implemented a Two-Proof Model using Proof-of-Integrity (PoI) and Proof-of-Offset (PoO) to provide verifiable project and offset lifecycle guarantees.

Privacy-Preserving Quantum-Secure Federated Learning Framework | *OpenFHE, TensorFlow/Keras* **Sep 2025**

- Built an end-to-end Federated Learning framework integrating Multi-Key CKKS Homomorphic Encryption with Proxy Re-Encryption using TensorFlow/Keras for secure distributed model aggregation across HPC nodes.
- Implemented encrypted communication pipelines using Apache Kafka and Mongoose HTTP server for secure model-weight exchange.

Parameter Optimisation & Feature Engineering for Time-Series ML on HPC **May 2025**

- Implemented and evaluated time-series forecasting models with systematic hyperparameter optimization and feature engineering for HPC environments, published the findings at SCI 2025.

Blockchain via Stealth Addresses: A PKI Approach **Nov 2024**

- Designed a PKI-based stealth address system for transaction privacy and unlinkability in decentralized networks.

Blockchain-Based E-Voting System **Jul 2021 – Jul 2022**

- Built a tamper-proof blockchain e-voting platform enabling secure and publicly verifiable remote voting.

Publications

- **Parameter Optimization and Feature Engineering for Time-Series Machine Learning Models on HPC** – 2025. Supercomputing India (SCI 2025).
- **Post Quantum Privacy Preserving Federated Learning with Multi-Key CKKS Homomorphic Encryption** – Manuscript under submission.

Education

B.Tech. in Computer Science & Engineering – 9.06 CGPA/10 **Jul 2018 – Aug 2022**
Dr. Babasaheb Ambedkar Technological University (DBATU) *Lonere, Maharashtra*

XII – Computer Science – 62.15% **Apr 2017 – May 2018**
Maharashtra State Board of Secondary & Higher Secondary Education (MSBSHSE)

X – Science – 9.8 CGPA/10 **Apr 2015 – May 2016**
Central Board of Secondary Education (CBSE)

Certifications & Training

Coursera, Microsoft: Advanced Data Analysis with Generative AI May 2026
CDAC Pune: Officer Training – Basic on Artificial Intelligence Jan 2026
Qualcomm Academy: AI Upskilling – Technical Foundation Jan 2026
CDAC Hyderabad: Machine Learning for Cyber Security Oct 2025
Coursera, Google: Agile PM, Git & GitHub, Computer Networking, Crash Course Python 2023
Coursera, IBM: Blockchain Essentials, Intro to DevOps, Data Analysis with Python 2022–2023
Coursera: Linux Fundamentals, Python Programming 2023